LESSON PLAN

BASIC RADIO & AVIONICS (25)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Brief description of engine starter motor, electric starter motor, cartridge turbo starter. Types of starting system, auxiliary power unit, air starter and battery trolleys Purpose and principle of operation of high energy ignition unit
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Brief description of engine starter motor, electric starter motor, cartridge turbo starter. Types of starting system, auxiliary power unit, air starter and battery trolleys Purpose and principle of operation of high energy ignition unit
5. Specific Objectives	25.1.2.1 Trainee will be able to know and understand the constructional details of starters, ignition units and principles of. operation of various engine instruments

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of starter motor diagrams, identification of components on models, safety precautions regarding high voltage and compressed air, instructor supervision during component identification
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an engine starter motor?
(b) Name two types of starting systems used in aircraft.
(c) What is the purpose of an Auxiliary Power Unit (APU)?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Constructional details of engine starter motor • Working principle of electric starter motor • Operation of cartridge turbo starter • Types of starting systems overview • Components of auxiliary power unit	L & Ds	PPT, IB	15 min
(b)	• Air starter and battery trolleys operation • Purpose of battery trolleys in starting • High energy ignition unit principle • Construction of high energy ignition unit • Safety protocols for starting systems	L & Ds	PPT, PDF	20 min
(c)	• Comparison of starter motor types • Principles of operation of various engine instruments • Integration of ignition unit with starter • Troubleshooting common starter faults • Maintenance requirements for starting systems	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between an electric starter and a cartridge turbo starter.
(b) Describe the principle of operation of a high energy ignition unit.
(c) List three safety precautions when handling battery trolleys.

12. Home Assignment and Follow Up	Prepare a brief note on the types of starting systems and their applications, referencing LP 02 for further reading on engine instruments.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Principle of operation of various engine instruments, viz, torque meter, thrust meter, oil pressure gauges, fuel flow meter, jet pipe temperature indicator, throttle lever, Position indicator, tachometer etc. Introduction to concept of glass cockpit	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min